Differential effects of the vesicular stomatitus virus M protein on NF-κB activation in L-929 and HeLa cells

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Abstract

Previously, we reported that the vesicular stomatitis virus (VSV) matrix (M) protein regulates NF-κB in mouse L929 cells. NF-κB remained cytoplasmic in wt-infected cells during the times tested (up to 4 hours post-infection, hpi). In contrast, NF-κB was rapidly activated between 2 and 2.5 hpi (20% and 87% nuclear localization respectively) in L929 cells infected with the T1026R1 (R1) mutant. This virus contains the M51R M protein mutation, which reduces the ability of M protein to inhibit host gene expression. Using qPCR analysis we determined that very little to no IFN-β mRNA is produced in wt infected cells, while significant amounts of IFN-β mRNA were produced in R1-infected cells. These findings suggest that VSV regulates the IFN pathway, at least in part, by limiting a step upstream of NF-κB activation in L929 cells.Our recent immunofluorescence studies in HeLa cells indicate that NF-κB is regulated in a cell-specific manner. In contrast to our findings in L929 cells, NF-κB remained cytoplasmic in R1-infected HeLa cells. Even after 10 hours of infection at an MOI of 10, NF-κB was found in the nucleus of only 20% of R1-infected cells. By 4 hours post-infection (hpi) at an MOI of 10, NF-κB localized to the nucleus of approximately 15% of wt-infected HeLa cells. This activation steadily increased to approximately 70% nuclear localization by 10 hpi. Therefore, the kinetics of NF-κB activation varied; while rapid activation was seen at early times post-infection in R1-infected L929 cells, a slow and steady activation occurred in wt-infected HeLa cells. Taken together, these results indicate that VSV regulates NF-κB in a cell-type specific manner.

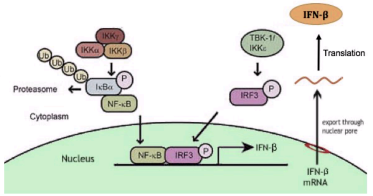


Image I. Role of NF-κB in the IFN-β expression pathway.

Previous Work

Differential expression of IFN-β mRNA in VSV-infected L929 and HeLa Cells

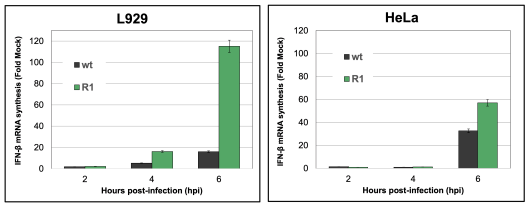


Figure 1. IFN-β mRNA production in L929 (A) and HeLa (B) cells infected with VSV. Cells were infected with either WT or R1 VSV at an MOI of 5. RNA was isolated from whole cells at the indicated times post-infection. IFN-β mRNA production in each sample was quantified by real-time qPCR (Taqman, ABI). Data are expressed as fold change relative to mock infected samples. Levels of the endogenous control HPRT were stable over the time course tested.

Immunofluorescence studies of NF-kB Activation in VSV-infected L929 cells

NF-κB is rapidly activated in L929 cells infected with strains of VSV encoding a M51R mutation in the M protein

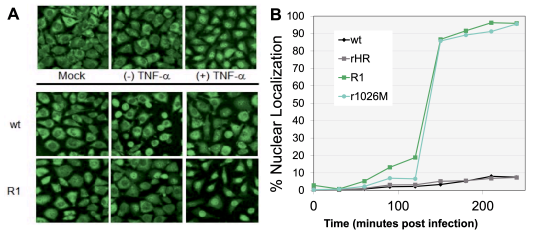
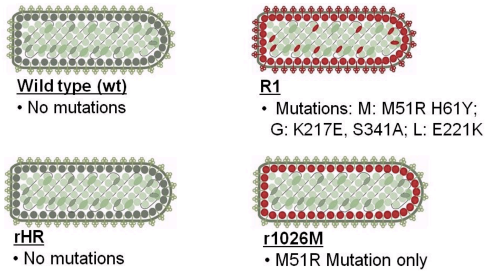


Figure 2. Percent nuclear localization of NF-κB in VSV-infected L929 cells. Cells were infected with a VSV strain that encodes a wt M protein (wt, rHR) or M-defective virus (R1, r1026M) at an multiplicity of infection (MOI) of 10 for the indicated time postinfection. Localization of the p65 subunit of NF-κB was visualized by immunofluorescence and confocal microscopy. (A) The percent nuclear localization was determined by counting the number of cells containing NF-kB associated fluorescence in their nuclei relative to the total number of cells imaged. (B) The data represent the mean of four independent experiments.

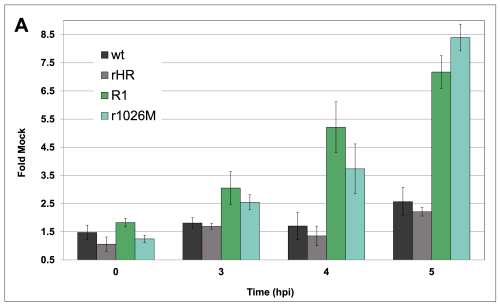
Viruses Used



Results

Effect of M mutants on NF-κB Activation

NF-κB Activation is delayed in L929 cells infected with viruses encoding a wt-M protein



NF-κB is Activated in HeLa cells infected with viruses encoding a wt-M protein

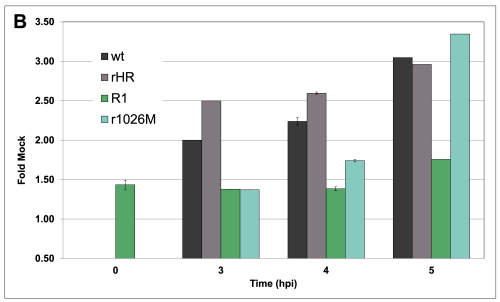


Figure 3. NF-κB activation in VSV-Infected Cells. (A) L929 or (B) Hela cells were infected at an MOI of 5 for the indicated hours post infection (hpi) and nuclear extracts were prepared using the Nuclear Isolation Kit from Active Motif. DNA-binding activity of NF-κB (p65) was determined using the ELISA-based TransAM Assay (Active Motif). Data presented as fold mock. Data represent means of two duplicates. The experiment was carried out four times in L929 cells and one time in HeLa cells.

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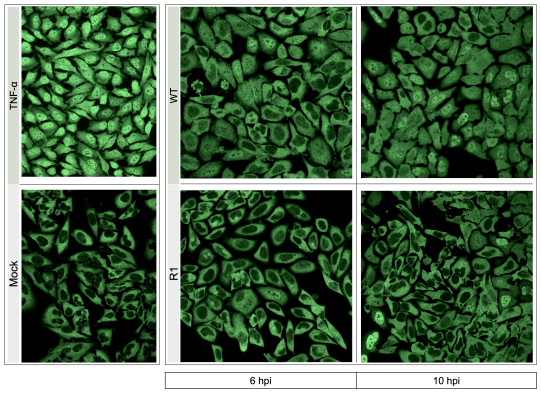
NF-kB Nuclear Translocation in VSV-Infected HeLa cells

A. Visualization of NF-kB localization in Infected HeLa Cells by Confocal Microscopy



B. Comparative analysis of NF-kB Activation kinetics in L929 and HeLa cells

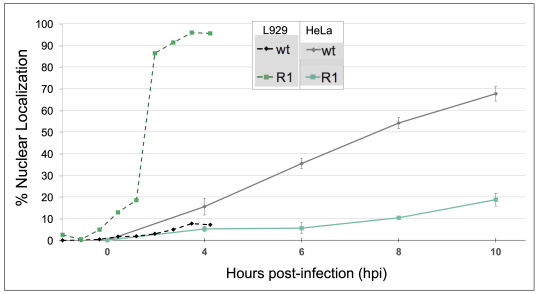


Figure 4. HeLa cells were grown on glass coverslips and infected with either wt or R1 VSV at an MOI of 10 PFU/cell. Cells were fixed at 4, 6, 6, and 10 hpi, and the coverslips were subsequently treated with the anti-NF-kB p65 AlexaFluor 488 conjugated antibody from SCBT. Cells were imaged with the Leica TCS SP5 Confocal using standard parameters for AlexaFluor 488. (A) Percent nuclear localization in HeLa cells was determined as stated in Fig. 1 legend. The resulting graph was rendered underneath the data from experiments done in L929 for the sake of direct visual comparison. (B)

Summary

	Virus	Mutation in M	Host Transcription Inhibited	NF-kB Activation by IF	NF-kB Activation by Trans-AM	IFN mRNA
L929	wt	None	Yes	No	No	No
	R1	M51R H61Y	No	Yes	Yes	Yes
HeLa	wt	None	Moderately	Yes 6hpi	Yes	Moderate
	R1	M51R H61Y	No	Moderate 8hpi	Moderate	Yes

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